# Vansh

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## Education

**B.Tech in Computer Science Engineering, Vellore Institute of Technology** *Sep 2023 – Expected Aug 2027* | ***8.81 CGPA***
**Class XII**, Floradale School, Ganganagar, Rajasthan ***84.8%***
**Class X**, D.A.V. Public School, Suratgarh, Rajasthan ***95.33%***

## Technical Skills

**Languages**: Java, Python, C++, JavaScript, TypeScript
**Web Technologies & Libraries**: React, Node.js, Express.js, Next.js, HTML/CSS, WebSockets, JWT
**Databases**: PostgreSQL (Supabase), NoSQL (MongoDB), MySQL, Redis (Upstash), Prisma
**Cloud & DevOps**: AWS, Docker, CI/CD (GitHub Actions), Vercel, Render
**Tools & Testing**: Git, GitHub, Playwright
**Core Concepts**: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs

## Experience

**Software Development Engineer Intern — Digi Marking Infotech Pvt Ltd** *Jun 2026 – Aug 2026 | Hybrid*

- Validated **UI components** and **30+ REST API payloads** across **3 client projects** to resolve edge cases, authoring **11+ bug reports** and system documentation to standardize workflows.
- Refactored **10+ complex monolithic frontend files** by isolating state logic into **25+ reusable UI components**, reducing code redundancy and significantly improving **debugging speed**.
- Automated **40+ end-to-end tests** with **Playwright** covering **authentication workflows** and routing, then delivered the suite to DevOps for **CI pipeline integration** to prevent staging regressions.

**Web Development Intern — KL LAB** *May 2025 – Jul 2025 | Remote*

- Engineered secure and scalable **REST APIs** with **Node.js** and **Express.js**, adding **JWT authentication** that guarded **20+ endpoints** and cut unauthorized access attempts by **30%**.
- Modeled **MongoDB schemas** with **Mongoose**, speeding up data retrieval for user, blog, and comment workflows by **40%** and cutting query latency to **under 50ms**.
- Deployed backend systems to **cloud infrastructure**, collaborating with a 4-person frontend team to achieve seamless **full-stack integration** and maintain **99.9% uptime** post-launch.

**Technical Team Core Member — AWS Club, VIT Bhopal** *Oct 2024 – Jul 2025 | Bhopal, MP*

- Orchestrated **5+ technical workshops** on **AWS fundamentals** for **100+ members**, driving a **20% increase** in club engagement and leading team-based **cloud architecture projects**.

## Projects

**Agentic Test Case Generator** | **GitHub** | **Live Demo** ***TypeScript, Node.js, Express.js, Next.js, Playwright, PostgreSQL, Docker***

- Developed an automated web testing platform that turns public web URLs into runnable, downloadable **Playwright E2E test suites** with browser trace replays, using **dual-model LLM planning** and **SHA-256 caching** to cut repeat-run LLM token usage by **85.5%**.
- Compiled structured JSON plans into typed **Playwright TypeScript** with a **3-round self-correction loop** for failing locators, reaching a **79.7%** pass rate across **522 generated tests**, which **lowered hallucinated LLM test scripts**.
- Ran the stateful Express job queue and SSE stream on **Render (512 MB)** for the **Next.js UI** while offloading headless Chromium over authenticated WebSockets to scale-to-zero **Google Cloud Run** to prevent OOMs, backed by **Supabase PostgreSQL**.
- Secured the test runner against unsafe generated code and leaked logins by blocking dangerous globals (`eval/process/fetch`) pre-execution, **stripping secrets from runner subprocesses**, and **masking user credentials**, verified across **19 TypeScript test files** in **GitHub Actions CI**.

**AI-Powered ATS Resume Screener** | **GitHub** | **Live Demo** ***Next.js, React, Shadcn, Redis, Prisma, PostgreSQL, TypeScript, LLMs***

- Architected an **enterprise ATS simulation platform** that evaluates resumes concurrently against **traditional** and **modern** applicant tracking systems, using distinct parser strictness and **semantic evaluation rules**.
- Reduced API latency and token consumption by **50%** via **LLM prompt consolidation** and implemented **SHA-256 normalized response caching** via **Upstash Redis**, reducing repeated scan latency by **97%** to **under 1 second**.
- Built a **high-availability inference pipeline** cascading from **Gemini models** to **Groq 120B** and **Groq 20B** models with **exponential back-off** and a **three-tier sliding window rate limiter** (Redis - PostgreSQL - fail-open), guaranteeing near **100% uptime** during provider rate limits.
- Designed a highly concurrent data architecture featuring a fail-open **Prisma 7 rate-limiting system** and unified **PostgreSQL logging transactions**, deployed behind edge-protected **Clerk v6 middleware**.

## Certifications

- AWS Cloud Practitioner Essentials - Nasscom